

RYAN LAU JUN HONG

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EDUCATION

Asia Pacific University (APU)

BS in Software Engineering | Feb 2026 - Present

(CGPA: / 4.0)

Diploma in ICT | Aug 2023 – Aug 2025

(CGPA: 3.56 / 4.0)

SKILLS & TECHNOLOGIES

Languages: TypeScript, Go, Dart, Python, Java

Frameworks: React.js, Next.js, TanStack Start, React Native, Flutter, Node.js

Tools: Docker, Git, Turborepo, PostgreSQL, MySQL, Redis, Keycloak (OIDC), GitHub Actions, Linux

WORK EXPERIENCE

Software Engineer Internship

First Pavilion Technology Sdn Bhd

September 2025 – January 2026

- **Owned the end-to-end design of a centralized exchange backend** in Go, implementing core wallet flows and tiered KYC/CDD compliance screening to handle high-throughput financial transactions safely.
- **Spearheaded the technical design of a real-time event pipeline** using Redis Pub/Sub and Server-Sent Events (SSE) to trigger frontend React Query invalidations, optimizing data synchronization latency.
- **Upgraded user authentication and Identity Management (IAM)** infrastructure by integrating Keycloak for centralized OIDC protocols and managing version-controlled database migrations via Atlas.
- **Built and automated the CI/CD lifecycle** using GitHub Actions to automate testing and deployment of Dockerized services to a Contabo VPS managed via aaPanel.

PROJECT EXPERIENCE

LLM-Wiki | AI Knowledge Retrieval System

[GitHub Repo](#)

- **Orchestrated a TypeScript monorepo** using Turborepo and automated Biome for automated code quality workflows, executing strict type-checking, linting and caching across decoupled frontend and API layers.
- **Designed a hybrid search Retrieval-Augmented Generation (RAG) pipeline** integrating Gemini and local Ollama models achieving sub-200ms context-retrieval latency during local benchmarking.
- **Built a real-time markdown parser and file watcher** to continuously synchronize local Obsidian vaults into a PostgreSQL database, utilizing Drizzle ORM for strict type safety.

LedgerTrace | Middleware Analytical Project

[Live Demo](#) | [GitHub Repo](#)

- **Developed an asynchronous middleware pipeline** using BullMQ and Redis to ingest Stripe and TRON events, load-tested to process simulated high-frequency payloads with real-time FX enrichment.
- **Enforce strict data privacy and infrastructure by deploying a TEE** (Trusted Execution Environment) via Chutes AI to evaluate inbound transactions safely against active invoices.
- **Implemented deterministic policy gating** to automatically post validated transactions to a Blk double-entry ledger while routing low-confidence anomalies to an exception desk for human review.

LANGUAGES

English: Proficient | **Chinese:** Native | **Malay:** Fluent